

BY 537: Computational Biology-1

Course Code	BY 537
Course Name	Computational Biology-1
L-T-P-C	3-0-2-4
Prerequisites	IC136 - Understanding Biotechnology & its Applications or Consent of faculty member
Intended for	Core for M.Tech Biotechnology and elective for B.Tech, M.Sc/M.Tech and PhD candidates
Approval	42nd Senate (Jan 2024), in the M.Tech Biotechnology (Systems Biology specialization) curriculum; 54th BoA (May 2024)

Course Contents

- **Module-1:** Introduction to Basic Programming Introduction to basic scripting and programming routinely used in computational biology. (xx hours)
- **Module-2:** Biological Databases and Sequence File Formats Introduction to different biological databases, their classification schemes, and biological database retrieval systems. (xx hours)
- **Module-3:** Sequence Alignments Introduction to concept of alignment, Scoring matrices, Alignment algorithms for pairs of sequences, Multiple sequence alignment. (xx hours)
- **Module-4:** Gene Prediction Methods What is gene prediction? Computational methods of gene prediction-prokaryotic & eukaryotic. (xx hours)
- **Module-5:** Molecular Phylogeny Introduction to phenotypic and molecular phylogeny. Representation of phylogeny, Molecular clocks, Methods of phylogenetic construction, statistical evaluation of the obtained phylogenetic trees. (xx hours)
- **Module-6:** Introduction to systems biology Different Omics, Metabolic pathways and networks. (xx hours)

Text Books

1. Dekking, F.M., Kraaikamp, C., Lopuhaa, H.P., Meester, L.E., **A Modern Introduction to Probability and Statistics - Understanding Why and How**, Springer.
2. S.C. Rastogi, N. Mendiratta, P. Rastogi, **Bioinformatics: Methods and Applications Genomics, Proteomics, and Drug Discovery**, 3rd Edition, PHI Learning Private Limited, 2011.
3. Z. Ghosh and B. Mallick, **Bioinformatics Principles and Applications**, Oxford University Press.
4. Arthur M. Lesk, **Introduction to Bioinformatics**, 3rd Edition, Oxford University Press.
5. Latest research articles will be advised related to the topic being taught.
6. Norman Bailey, **Statistical methods in Biology**, latest edition.

References

1. NA